Theory Questions

a) Kubernetes Architecture

- **Kubernetes Cluster:**
  - control plane:
    - kube-apiserver: expose Kubernetes API server and handles communication between components of cluster
    - scheduler: assigns newly created pods with no nods to run on
    - control-manager: complie different controllers into 1 bin file (node controller, job controller, etc.)
    - etcd: key-value cluster data storage
  - node:
    - kublet: small agent to ensure containers are running in pods and communicate with apiserver
    - pods
- **Pods vs Containers**: pods are the smallest deployable units in k8s and can contain one or more containers

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b) Deployments & Services

- **Deployments**:
  - maintain desired state dynamic pods management
  - Ensures High Availability through:
    - Rolling updates: updates with zero down-time
    - Self-Healing: recreates pods if one crashes or goes down
    - Rollbacks: reverts to previous states if a deploy has a bug
- **Services**:
  - ClusterIP: expose svc on internal IP and accessed from within the cluster, used for internal services communication
  - NodePort: exposes svc to an external port on every node in the cluster, used to allow access for development/testing
  - LoadBalancer: assigned an external IP, receives traffic and distributes traffic to nodes in cluster exposing only one end-point

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c) Scaling & Autoscaling

- **K8s Scaling**:
  - Pod scaling: adjusts number of pods or allocated resources per pod
  - Node scaling: adds/removes pods from cluster based on resources demand
- **HPA**:
  - Updates workload resources according to metrics when it reaches a certain threshold to either:
    - **scale up**: scale horizontally when avg utilization of a metrics exceeds target metrics
    - **scale down**: decrease number of pods after traffic rush cooldown to save cluster resources